

Tutorial 01. Perspective Projection

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This week: Python programming and a projection demonstration. Use the separate Python notebook for programming practice and this sheet for geometric reasoning.

Interactive lab: <https://pinhole-camera-lab.chenrui-fan.chatgpt.site/>

Explore the model, predict what changes, and vote for the questions you would like to discuss. The instructor will release answers for individual reading and reflection before the class discussion.

Convention. The camera centre is O , $f > 0$, and scene points lie at $Z > 0$ unless stated otherwise. Use the *virtual* image plane, principal point at the origin, and image axes parallel to X, Y , with y upward:

$$\pi(X, Y, Z) = (x, y) = \left(f \frac{X}{Z}, f \frac{Y}{Z}\right).$$

The question illustrations use $f = 1$. The main demo's physical sensor lies behind the pinhole: displacements from its principal point use the opposite projection sign.

Core questions – the numbers match Q01–Q07 on the website.

- Q1.** Show that the projectable points of a straight 3D line are collinear in the image. State what happens for a line through the camera centre.
- Q2.** If two 3D lines share a projectable point, show that its image belongs to both projected sets. Must the two image lines have a unique crossing?
- Q3.** Show by a concrete counterexample that intersecting images do not imply that the corresponding 3D lines intersect.
- Q4.** Show by a counterexample that angles are not preserved in general under perspective projection.
- Q5.** Show that lengths are not preserved in general. State the assumptions under which a vertical segment has image height $h = fH/Z$.
- Q6.** For a level camera, determine the 3D plane through the camera centre that corresponds to the horizon. Distinguish this plane from the actual ground below the camera.
- Q7.** Show that parallel 3D lines may converge to a vanishing point in the image. Derive the point for direction V when $V_z \neq 0$, and discuss $V_z = 0$.

Optional extensions – not in the website question bank.

- E1.** Determine the perspective silhouette of an opaque sphere of radius $R > 0$ and centre $S = (S_x, S_y, S_z)$, assuming $S_z > R$ so the entire sphere is in front of the camera. When is its image a circle?
- E2.** Under orthographic projection along Z , what is the image of the same sphere?

Topic numbering: Q1–Q6 retain their numbers; old Q9 is now Q7; old Q7 and Q8 are now E1 and E2.